BAMAT207: APPLICATIONS OF PROBABILITY AND STATISTICS

PROJECT ACTIVITY REPORT: PROBABILISTIC SMS SPAM DETECTION

Course Name & Code:	Applications of Probability and Statistics (BAMAT207)
Topic:	Probability and Random Variables
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GitHub Repository Link:	https://github.com/Harshith1987/BAMAT207-Probabilistic-SMS-Spam-Detection

1. Executive Summary & Problem Formulation

Unsolicited mobile text messages (SMS spam) present critical challenges in digital communications, including fraud, phishing, and malware distribution. The objective of this project is to develop a mathematically rigorous probabilistic classification system that categorizes an incoming SMS message as either Spam (Y = 1) or Genuine / Ham (Y = 0).

The project applies foundational principles of Probability and Random Variables: computing prior probabilities, evaluating word-level empirical conditional probabilities, mathematically proving and verifying Bayes' Theorem, formulating the classification process as a discrete Bernoulli Random Variable (calculating Expectation and Variance), training a Multinomial Naive Bayes classifier with Laplace smoothing, and conducting an in-depth confusion matrix error audit of False Positives and False Negatives.

2. Dataset Handling & Prior Probabilities

The project uses SMS Spam Collection dataset from the UCI Machine Learning Repository (ID: 228). Data hygiene was enforced by validating null values and removing duplicate entries to prevent estimation bias.

Metric / Parameter	Value	Statistical Interpretation
Raw Messages	5,572	Original corpus from UCI repository
Missing / Null Values	0	No data imputation required
Duplicate Records Removed	403	Eliminated redundant messages
Clean Unique Messages (N)	5,169	Sample space size for probability analysis
Prior Probability P(Ham)	0.8737 (4,516 msgs)	87.37% baseline probability of genuine SMS

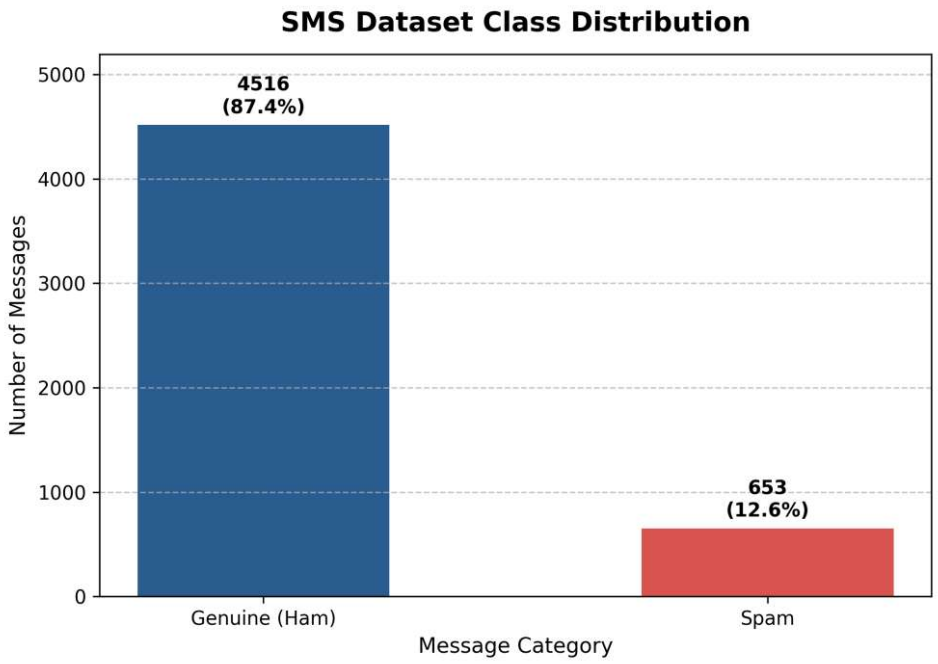


Figure 1: SMS Dataset Class Distribution (Ham vs Spam)

3. Word-Based Probability Analysis & Bayes' Theorem Verification

For any suspicious word W, we calculate the conditional probability P(Spam | W) via two independent methods:

1. Empirical Ratio: P(Spam | W) = Count(Spam containing W) / Count(All messages containing W)
2. Bayes' Theorem: P(Spam | W) = [P(W | Spam) * P(Spam)] / P(W)

Both formulations produce identical numerical results, demonstrating the theorem in practice.

Word (W)	Total (N_W)	Spam (N_S)	P(W|Spam)	P(Spam|W) Bayes	Equivalence Verified
free	203	148	0.2266	0.7291 (72.91%)	True
prize	73	73	0.1118	1.0000 (100.0%)	True
win	55	49	0.0750	0.8909 (89.09%)	True
urgent	62	57	0.0873	0.9194 (91.94%)	True
offer	30	24	0.0368	0.8000 (80.00%)	True

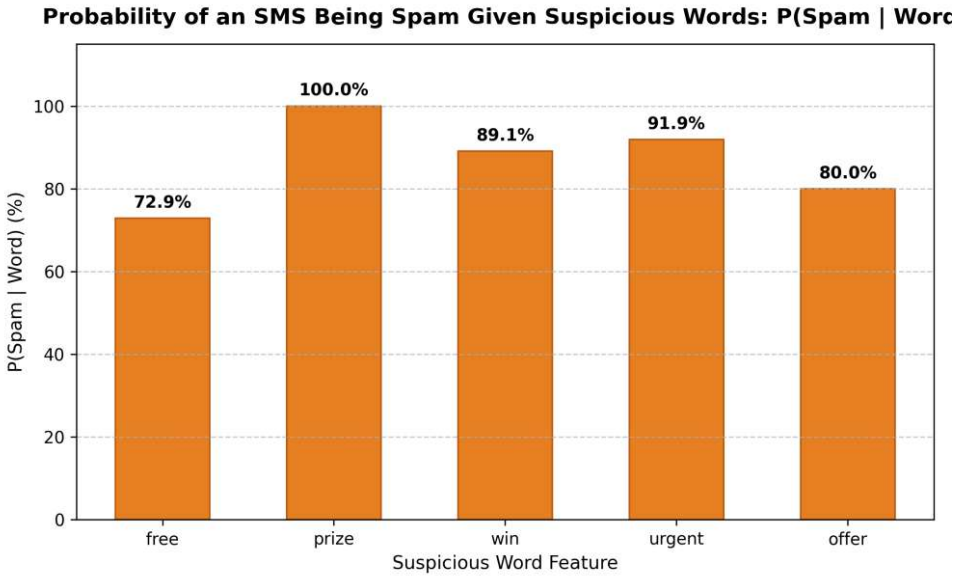


Figure 2: Conditional Probability of Spam Given Suspicious Words: P(Spam | Word)

4. Random Variable Analysis (Bernoulli Distribution)

Let the random variable Y define the classification state of an arbitrary message:
Y = 1 if the message is Spam
Y = 0 if the message is Ham

Here, Y follows a Bernoulli distribution: Y ~ Bernoulli(p) where p = P(Spam) = 0.1263 and q = 1 - p = 0.8737.

• Expectation E(Y):
E(Y) = sum(y * P(Y = y)) = (1 * p) + (0 * q) = p = 0.1263

• Variance Var(Y):
Var(Y) = E[Y^2] - (E[Y])^2 = p - p^2 = p(1 - p) = 0.1263 * 0.8737 = 0.1104

• Standard Deviation SD(Y):
SD(Y) = sqrt(Var(Y)) = sqrt(0.1104) = 0.3322

5. Model Building: Multinomial Naive Bayes

• Data Partitioning: 80% training set (4,135 samples) and 20% testing set (1,034 samples) using stratified sampling (stratify=y) to maintain identical class proportions.
• Feature Extraction: CountVectorizer extracted 7,414 unique word tokens as features.
• Classification Algorithm: Multinomial Naive Bayes (MultinomialNB) with Laplace Add-1 smoothing (alpha = 1.0).

6. Model Evaluation & Error Analysis

Evaluation Metric	Model Score
Overall Test Accuracy	98.36% (0.9836)
Precision (Spam)	95.97% (0.9597)
Recall (Spam)	90.84% (0.9084)
F1-Score (Spam)	0.9333

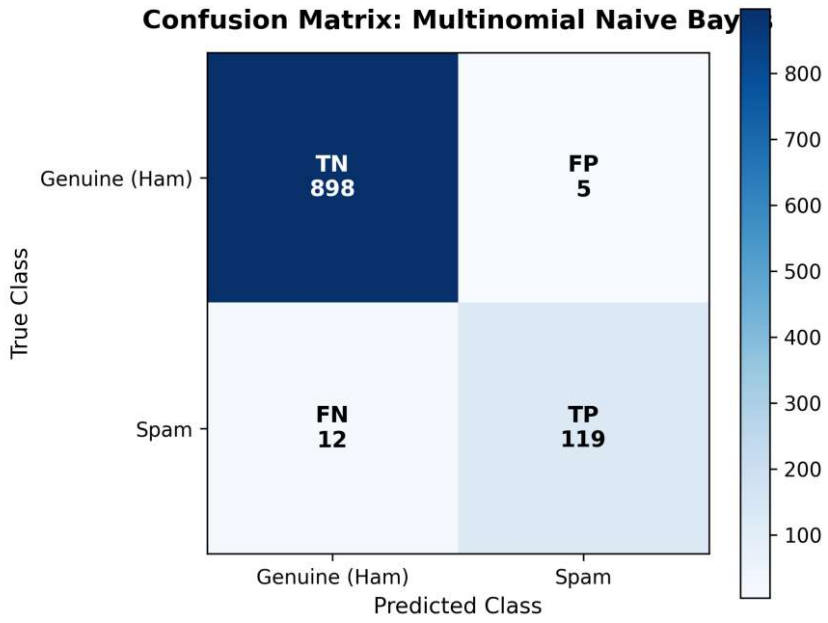


Figure 3: Confusion Matrix (Multinomial Naive Bayes on 1,034 Test Samples)

Critical Error Audit:

1. False Positives (Genuine messages classified as Spam): 5 messages (0.55% error rate on Ham). High precision is paramount to ensure legitimate personal or business messages are not filtered out.
2. False Negatives (Spam messages classified as Genuine): 12 messages.

7. Real-Time Testing & Inference Demonstrations

Input SMS Message	Prediction	P(Spam|Msg)	P(Ham|Msg)
Congratulations! You have won a free prize. Claim now.	Spam	1.0000 (100.0%)	0.0000 (0.0%)
Hey, are you free this afternoon to study for the	Ham	0.0006 (0.06%)	0.9994 (99.94%)

BAMAT207 exam?			
URGENT! Your mobile account has won an exclusive $1,000 cash offer.	Spam	1.0000 (100.0%)	0.0000 (0.0%)
Can you send me the lecture notes when you get home?	Ham	0.0028 (0.28%)	0.9972 (99.72%)
Win a brand new car today! Free entry, text WIN to 80085 immediately.	Spam	0.9999 (99.99%)	0.0001 (0.01%)

8. Conclusion & Learning Outcomes

1. Integrated foundational probability theory, conditional probability, Bayes' Theorem, and random variables into a functional AI classifier.
2. Demonstrated that the Bernoulli random variable variance is driven by class balance, and verified Bayes' theorem both analytically and empirically.
3. Evaluated a complete NLP pipeline achieving 98.36% accuracy and 95.97% precision, establishing practical trade-offs between precision and recall.
4. Established a professional GitHub repository profile for coursework submission: https://github.com/Harshith1987/BAMAT207-Probabilistic-SMS-Spam-Detection